

Research Project Proposal: Enhancing FedALA in the OpenFGL Framework

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1 Selected Result from OpenFGL to Reproduce

For this project, we will reproduce the federated learning performance results reported in **Table 7** of the OpenFGL paper [1]. Table 7 compares multiple FL algorithms on standard graph datasets under the OpenFGL setup, including:

- FedAvg [3]
- FedProx [4]
- GCFL
- FedGKT
- Local Training (LT)
- Global Full Graph Training

We will faithfully replicate the Table 7 experimental pipeline and datasets (e.g., Cora, Citeseer, Pubmed, ogbn-arxiv). This provides a baseline for evaluating both the original FedALA method and our improved version.

2 Selected Method to Extend: FedALA

We choose to extend **FedALA (Adaptive Layer Aggregation)** [2], a personalized FL method that assigns per-layer adaptive aggregation weights based on layer usefulness.

OpenFGL currently supports FedAvg, FedProx, GCFL, and several other FL strategies, but it does not include FedALA. Our first objective is to integrate FedALA into OpenFGL and reproduce its results under the same environment as Table 7.

3 Proposed Improvements to FedALA

To further improve FedALA performance on non-IID graph data, we propose four enhancements ranging from stability adjustments to initialization strategies.

3.1 Improvement 1: Attention Smoothing (FedALA-S)

(Primary improvement – focuses on weight stability)

Motivation

Original FedALA computes attention weights for each layer as:

$$\alpha_k^l(t) = \text{softmax}_k(s_k^l(t)) \quad (1)$$

However, on graph datasets, the attention weights **oscillate heavily** between rounds due to noisy local updates and heterogeneous node distributions. This instability harms convergence and increases variance.

Proposed Method

We introduce **temporal smoothing** to stabilize attention weights:

$$\alpha_k^l(t) = (1 - \beta)\text{softmax}(s_k^l(t)) + \beta\alpha_k^l(t - 1) \quad (2)$$

Where $0.7 \leq \beta \leq 0.95$ controls the smoothness.

Expected Outcomes

- Reduces round-to-round oscillation.
- More stable aggregation and faster convergence.
- Very low computational overhead ($< 1\%$).

3.2 Improvement 2: FedProx Regularization (FedALA-S+Prox)

(Secondary improvement – reduces client drift)

Motivation

FedALA does not explicitly control how far local models drift from the global model. On graph data, drift increases due to uneven degree distributions.

Proposed Method

We integrate a FedProx proximal term into each client’s local optimization:

$$\mathcal{L}_{prox} = \mathcal{L}_{local} + \frac{\mu}{2}\|W - W_{\text{global}}\|^2 \quad (3)$$

Where $0.001 \leq \mu \leq 0.05$ limits excessive drift.

3.3 Improvement 3: Momentum Integration for Update Direction

(Enhancement for update stability)

Motivation

In the standard FedALA, the initialized local model is constructed using the difference between the global model and the previous local model: $(\Theta^{t-1} - \Theta_i^{t-1})$. In highly non-IID graph settings, this instantaneous difference vector can be noisy and erratic, leading to unstable learning trajectories.

Proposed Method

We propose integrating a momentum term into the ALA update process. Instead of using only the current difference, we maintain a moving average of the update direction. Let V_i^t denote the momentum buffer for client i at round t :

$$V_i^t = \gamma V_i^{t-1} + (1 - \gamma)(\Theta^{t-1} - \Theta_i^{t-1}) \quad (4)$$

The adaptive aggregation is then applied to this smoothed difference vector rather than the raw instantaneous difference:

$$\hat{\Theta}_i^t = \Theta_i^{t-1} + V_i^t \odot \mathbf{W}_i^t \quad (5)$$

Where $\gamma \in [0, 1)$ is the momentum coefficient and $\mathbf{W}_i^t$ are the learned aggregation weights.

Expected Outcomes

- Smooths out the noise in the global-to-local information transfer.
- Accelerates convergence by accumulating gradients in consistent directions.
- Improves robustness against outlier clients in the graph network.

3.4 Improvement 4: Meta-Learning Initialization for ALA Weights

(Enhancement for computational efficiency and performance)

Motivation

Standard FedALA typically re-initializes the aggregation weights $\mathbf{W}_i$ (usually to ones or randomly) at the beginning of each round's "Weight Learning Phase". This "cold start" forces the client to re-learn the optimal interpolation between global and local models from scratch every round, wasting computational resources and potentially getting stuck in suboptimal local minima.

Proposed Method

We treat the learning of aggregation weights $\mathbf{W}_i$ as a meta-learning task where the state is preserved across rounds. We initialize the weights for round t using the converged weights from round $t - 1$:

$$\mathbf{W}_{init}^t \leftarrow \mathbf{W}_{converged}^{t-1} \quad (6)$$

During the ALA phase, we perform only a few steps of gradient updates (fine-tuning) instead of a full training cycle:

$$\mathbf{W}_i^t \leftarrow \mathbf{W}_{init}^t - \eta \nabla_{\mathbf{W}} \mathcal{L}(\hat{\Theta}_i, \mathcal{D}_{val}) \quad (7)$$

Expected Outcomes

- **Faster Adaptation:** The model starts with a "personalized" prior knowledge of which layers/parameters to trust.
- **Reduced Computation:** Drastically reduces the number of epochs required for the weight learning phase (e.g., from 10 epochs to 1 epoch).
- **Performance Gain:** Avoids the instability of re-learning weights from scratch on small local datasets.

4 Evaluation Plan

We will extend the experimental table (Table 7 in OpenFGL) by adding the following rows:

1. **FedALA** (baseline reproduction)
2. **FedALA-S** (Smoothing only)
3. **FedALA-S+Prox** (Smoothing + Regularization)
4. **FedALA-Complete** (Smoothing + Prox + Momentum + Meta-Init)

Evaluation metrics will match OpenFGL standards:

- Node classification accuracy
- Convergence curves (Loss vs. Round)
- Client variance analysis (Robustness)
- Attention/Weight stability visualization
- Computational overhead (Time per round)

All tests will be performed on the same datasets (Cora, Citeseer, Pubmed, ogbn-arxiv) and partitioning used by the OpenFGL authors.

References

- [1] Li, X. et al. (2023). *OpenFGL: A Comprehensive Library for Federated Graph Learning*. NeurIPS 2023 Datasets and Benchmarks Track.
- [2] Zhang, J., Hua, Y., Wang, H., Song, T., Xue, Z., Ma, R., & Guan, H. (2023). *FedALA: Adaptive Local Aggregation for Personalized Federated Learning*. AAAI Conference on Artificial Intelligence.
- [3] McMahan, B., Moore, E., Ramage, D., Hampson, S., & y Arcas, B. A. (2017). *Communication-Efficient Learning of Deep Networks from Decentralized Data*. AIS-TATS.
- [4] Li, T., Sahu, A. K., Zaheer, M., Sanjabi, M., Talwalkar, A., & Smith, V. (2020). *Federated Optimization in Heterogeneous Networks*. MLSys.